

WEEK 6 POWER BI REPORT

Healthcare Dashboard Development and Data Model Analysis

Submitted By

Harini R – E25DA021

1. Introduction

Week 6 focuses on using the prepared healthcare data model to create an interactive Power BI report. The work includes checking table relationships, using measures, and presenting healthcare information through cards, charts, tables and slicers. The dashboard provides a visual summary of patients, billing amounts, medical conditions, age groups, insurance providers and admission dates.

2. Objectives

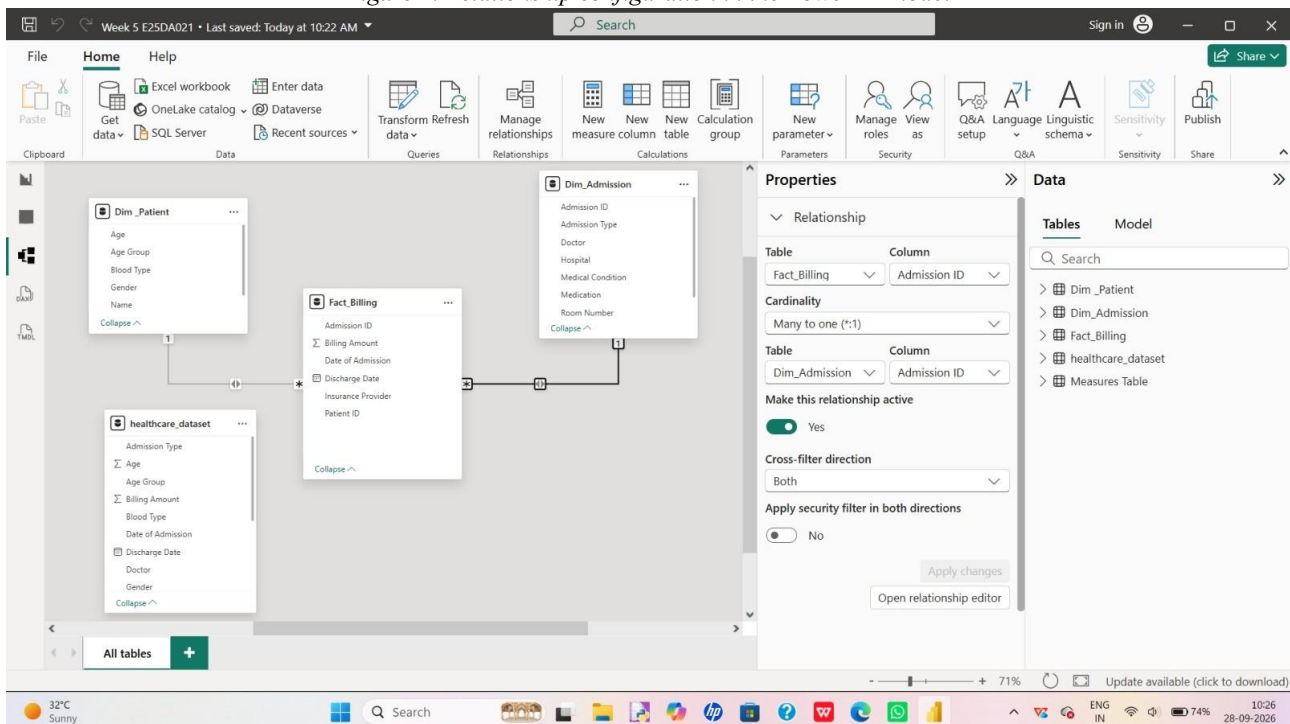
- To verify relationships between fact and dimension tables.
- To use the created DAX measures in report visuals.
- To display key healthcare indicators using card visuals.
- To analyze patient distribution by medical condition and age group.

- To compare total billing amounts across insurance providers.
- To provide interactive filtering using slicers.

3. Relationship Configuration

The relationship settings show Fact_Billing connected with Dim_Admission through Admission ID. The relationship is configured as Many-to-One, is active, and uses Both as the cross-filter direction. The model also connects patient information with billing information. These relationships allow fields from different tables to work together when creating report visuals.

Figure 1. Relationship configuration in the Power BI model

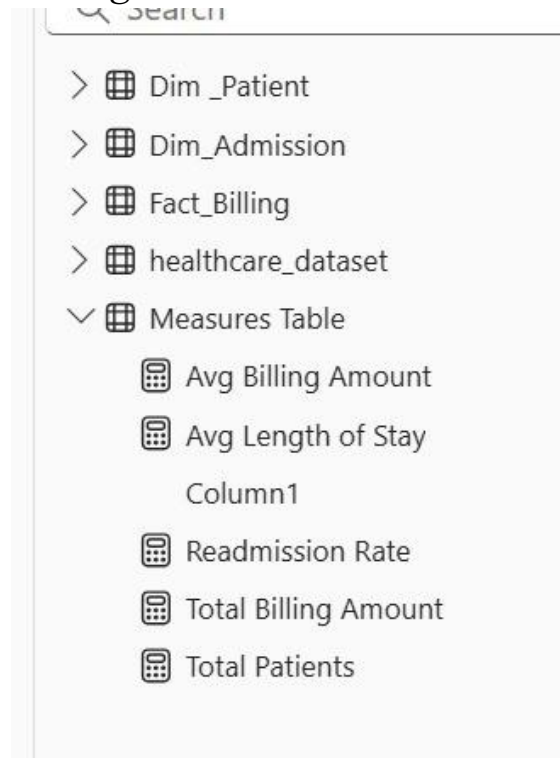


4. Measures Used in the Report

The Measures Table contains reusable calculations for the dashboard. The visible measures are Average Billing Amount, Average Length of Stay, Readmission Rate, Total Billing Amount and Total Patients. These measures

are used to display summarized values and support comparisons across different categories.

Figure 2. Measures Table



5. Key Performance Indicators

The dashboard uses card visuals to present important summary values. The displayed indicators include an average billing amount of 25.56K, an average length of stay of 15.50, a total billing amount of 1.28bn, and total patients of approximately 50K. These cards provide a quick overview of the dataset.

Figure 3. Dashboard KPI cards and patient analysis

25.56K

Avg Billing Amount

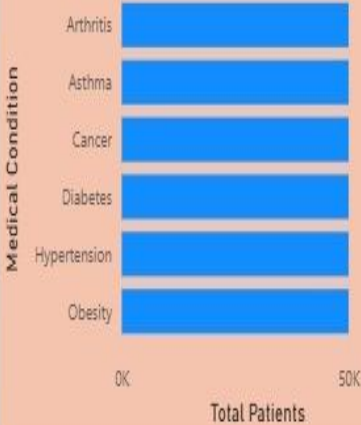
15.50

Avg Length of Stay

1.28bn

Total Billing Amount

Total Patients by Medical Condition

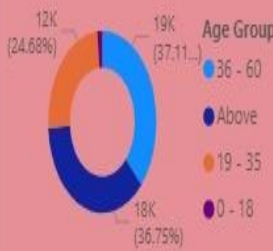


50K

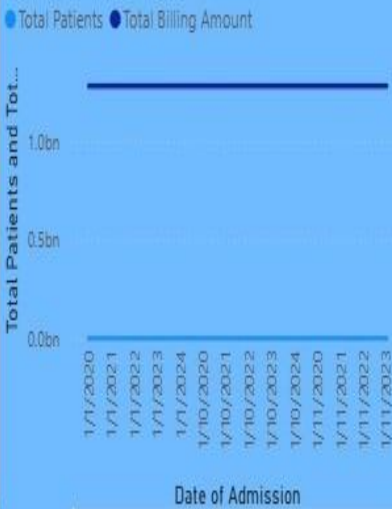
Total Patients

Medical Condition: Arthritis		
Age Group	Avg Billing Amount	Total
0 - 18	26,442.58	
19 - 35	25,116.56	
36 - 60	25,412.10	
Above	25,774.56	
Total	25,487.29	

Total Patients by Age Group



Total Patients and Total Billing Amount by Date of Admission



6. Patient Analysis by Medical Condition

A bar chart is used to show total patients by medical condition. The displayed categories include Arthritis, Asthma, Cancer, Diabetes, Hypertension and Obesity. The visual makes it easier to compare patient counts across the different medical conditions.

7. Patient Analysis by Age Group

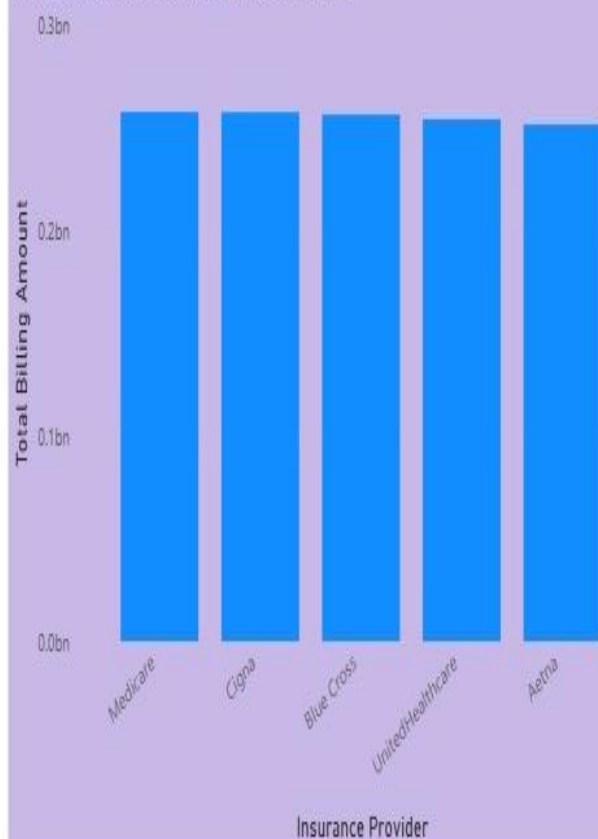
A donut chart presents the distribution of total patients across age groups such as 0–18, 19–35, 36–60 and Above. The dashboard also contains a table combining medical condition, age group and average billing information for more detailed analysis.

8. Billing Analysis by Insurance Provider

The dashboard includes a column chart showing Total Billing Amount by Insurance Provider. The providers displayed include Medicare, Cigna, Blue Cross, UnitedHealthcare and Aetna. This visual supports comparison of billing totals across insurance providers.

Figure 4. Billing analysis and slicers

Total Billing Amount by Insurance Provider



Insurance Provider

- ☐ Aetna
- ☐ Blue Cross
- ☐ Cigna
- ☐ Medicare
- ☐ UnitedHealthcare

Admission Type

- ☐ Elective
- ☐ Emergency
- ☐ Urgent

Date of Admission

- ☐ 1/1/2020
- ☐ 1/1/2021
- ☐ 1/1/2022
- ☐ 1/1/2023
- ☐ 1/1/2024
- ☐ 1/10/2020
- ☐ 1/10/2021

9. Interactive Slicers

Slicers are provided for Insurance Provider, Admission Type and Date of Admission. Users can select one or more values to filter the dashboard and examine specific portions of the healthcare data. This improves the interactive nature of the report.

10. Dashboard Analysis

Visual	Purpose
KPI Cards	Summarize average billing, average stay, total billing and total patients.
Medical Condition Chart	Compare patient counts across medical conditions.
Age Group Donut	Show patient distribution across age groups.
Billing by Insurance	Compare total billing amounts among insurance providers.
Date Trend	Examine total patients and billing amounts across admission dates.
Slicers	Filter the report by insurance provider, admission type and admission date.

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following columns: **Date of Admission**, **Billing Amount**, **Discharge Date**, and **Insurance Provider**. The table contains 21 rows of data. The interface includes a ribbon with various tools, a left sidebar with queries, and a right sidebar with query settings and applied steps.

	Date of Admission	Billing Amount	Discharge Date	Insurance Provider
1	1/31/2024	18856.28131	02-02-2024	Blue Cross
2	8/20/2019	33643.32729	26-08-2019	Medicare
3	9/22/2022	27955.09608	07-10-2022	Aetna
4	11/18/2020	37909.78241	18-12-2020	Medicare
5	9/19/2022	14238.31781	09-10-2022	Aetna
6	12/20/2023	48145.11095	24-12-2023	UnitedHealthcare
7	11/3/2020	19580.87234	15-11-2020	Medicare
8	12/28/2021	45820.46272	07-01-2022	Cigna
9	7/1/2020	50119.22279	14-07-2020	Cigna
10	5/23/2021	19784.63106	22-06-2021	UnitedHealthcare
11	4/19/2020	12576.79561	22-04-2020	Medicare
12	8/13/2023	7999.58688	05-09-2023	Medicare
13	12/12/2019	43282.28336	28-12-2019	Cigna
14	5/22/2020	33207.70663	19-06-2020	UnitedHealthcare
15	10/8/2021	40701.59923	13-10-2021	UnitedHealthcare
16	1/1/2023	12263.35743	11-01-2023	Blue Cross
17	6/23/2020	24499.8479	14-07-2020	UnitedHealthcare
18	3/8/2020	17440.46544	02-04-2020	Cigna
19	3/4/2021	18843.02302	14-03-2021	Blue Cross
20	11/15/2022	23762.20358	22-11-2022	Cigna
21	6/28/2023	42.51458855	02-07-2023	Blue Cross

11. Result / Outcome

An interactive healthcare analytics dashboard was developed using the prepared Power BI model. The report combines measures, relationships and multiple visualizations to summarize patient and billing information. Users can interact with the report through slicers and compare healthcare indicators from different perspectives.

12. Conclusion

Week 6 demonstrated how a structured Power BI data model can be converted into an interactive analytical dashboard. Relationships connected the tables, measures provided calculated values, and visuals presented the results in an understandable form. The completed report provides a useful foundation for further healthcare data analysis and dashboard development.